Model 2

Code for running Model 2, which calculates and saves the extinction time across the parameter space for a given species and time period.

Background Functions

Initialize functions for the Gaussian-Quadratic TPC and the logistic growth function. The timestep has been halved to accommodate the addition of 2 data points per day.

```
getParameters[species_String] :=
 Module[{row}, row = SelectFirst[speciesData, #[[1]] == species &];
  If[row === Missing["NotFound"], "Species not found", Rest[row]]]

r[T_] :=
  If[T ≤ Topt, Exp[-((T - Topt) / (2 σp)) ^ 2], 1 - ((T - Topt) / (Topt - CTmax)) ^ 2];

δt = .5;
```

$$nt[t\_, ntlast\_] := \frac{r[Z[[t]]]\ Exp[r[Z[[t]]] * \delta t]}{\left(\frac{r[Z[[t]]]}{ntlast} - \alpha\right) + \alpha\ Exp[r[Z[[t]]] * \delta t]}$$

Initialize functions for spectral synthesis and spectral mimicry, as well as a new function (‘changeautocorr’) that preserves seasonal cycling at each autocorrelation level.

```
cmplx[mod_, arg_] := ExpToTrig[mod Exp[I arg]];

SpecSynFourier[γ_, Nobs_, μ_ : 0, σ_ : 1, seed_ : 0] := Module[{
    phase, f, fseries, vec},
   If[seed == 0, SeedRandom[], SeedRandom[seed]];
   phase = RandomReal[{0, 2 Pi}, Nobs];
   f = Range[1 / Nobs, 0.5, 1 / Nobs];
   fseries =
    Join[{0 + 0 I}, Table[cmplx[1 / f[[i]] ^ (γ / 2 ), phase[[i]]], {i, 1, Length[f]}],
     Table[cmplx[1 / f[[i]] ^ (γ / 2 ), -1 * phase[[i]]], {i, Length[f] - 1, 1, -1}]];

   vec = InverseFourier[fseries, FourierParameters → {-1, 1}];
   Standardize[Re[vec[[1 ;; Nobs]]] * σ + μ];

changeautocorr[X_, γ_, samplesperyear_, seed_ : 0] := Module[{
   var, Y, fftdata, mean, mmeandevs, adjvar,
```

```
    powersp, int, lm, a, b, n = Length[X], phase, f, vec, Z},
   mean = Mean[X];
   var = Variance[X - mean];
   mmeandevs =
    Table[Mean[Table[X⟦i⟧, {i, j, n, samplesperyear}]], {j, 1, samplesperyear}];
   Y = Table[X⟦i⟧ - mmeandevs⟦Mod[i, samplesperyear, 1]⟧, {i, 1, n}];
   adjvar = Variance[Y];

   fftdata = Fourier[Y, FourierParameters → {-1, 1}];
   powersp = Table[{i / n, Abs[fftdata⟦i + 1⟧] ^ 2}, {i, 1, n / 2 - 1}];
   lm = LinearModelFit[{Log[#1], Log[#2]} & @@@ powersp, x, x];

   If[seed == 0, SeedRandom[], SeedRandom[seed]];
   phase = RandomReal[{0, 2 Pi}, n];
   f = Range[1 / n, 1, 1 / n];
   vec = InverseFourier[
     Join[{0 + 0 I}, Table[cmplx[1 / f⟦i⟧ ^ (γ / 2 ), phase⟦i⟧], {i, 1, Length[f]}],
      Table[cmplx[1 / f⟦i⟧ ^ (γ / 2 ), -1 * phase⟦i⟧], {i, Length[f] - 1, 1, -1}]],
     FourierParameters → {-1, 1}];
   Z = Standardize[Re[vec]] * Sqrt[adjvar];

   {{lm, var, adjvar}, Table[Z⟦i⟧ + mmeandevs⟦Mod[i, samplesperyear, 1]⟧, {i, 1, n}]}
  ]

Mimic1[x_, z_] := Module[{y},
    y = x;
    Do [y⟦Ordering[z, i]⟦i⟧⟧ = x⟦i⟧, {i, 1, Length[x]}];
    y];
```

Import Species Data

Imports the data needed for simulations. Manually select which species and time period you want before each run.

```
species = "T.citricidus";(*select species. watch out for spelling!*)
when = "1994-01-01to2003-12-31";
(*1994-01-01to2003-12-31 OR 2014-01-01to2023-12-31*)

importfrom =
  "/Users/alisonrobey/Documents/PhD Year 3 (2023-2024)/Ch1/Code/Model3/";
TPCdata = Import[importfrom <> "speciesparams.xlsx"];
speciesData = TPCdata⟦1, 2 ;;⟧;

parameters = getParameters[species]
CTmax = parameters⟦1⟧;
Topt = parameters⟦2⟧;
σp = parameters⟦3⟧;
lat = ToString[parameters⟦4⟧];
long = ToString[parameters⟦5⟧];

data = Import[importfrom <> "VisualCrossingData/" <>
    species <> "_" <> lat <> "," <> long <> "_" <> when <> ".csv", "CSV"];
temps = Flatten[data⟦2 ;;, {2, 3}⟧];
pairedTemps = Partition[temps, 2];
meanTemps = Mean /@ pairedTemps;
```

Out[ ]=

```
{30.92, 26.4, 4.23, 34.98, 138.4, 0.392489, 0.522572, 0.415541,
 0.839454, 16.4832, 9.0819, 17.5192, 8.48219, Toxoptera citricidus, NHEX}
```

Run Simulation

Runs simulations and stores extinction times (and, if desired, minimum population sizes, though these sections are currently commented out). Set parameters for size of run and export location. Full sized parameter runs used in final results are shown in parenthesizes. If extinction threshold is changed, output file names need to be changed manually.

```
(*simulation params*)
α = .001;
tmax = Length[temps];
maxsims = 10; (*1000*)
γvals = Range[0, 2, .5]; (*Autocorrelation: Range[0,2,.1] → 21*)

(*sort paired temps by their mean*)
W = SortBy[pairedTemps, Mean[{#〚1〛, #〚2〛}] &];

(*set initial pop size to mean pop size for each parameter set*)
ninit = Mean[Table[r[temps〚i〛], {i, 1, Length[temps]}]] / α;

(*initialize array to store extinction times; =tmax if not extinct*)
extTime = ConstantArray[tmax, {Length[γvals], maxsims}];
(*minPop=ConstantArray[ninit,{Length[γvals],maxsims}];*)

Do[γ = γvals〚i〛;

  X = changeautocorr[meanTemps, γ, 365, 0];
  Z = Mimic1[W, X〚2〛];
  Z = Flatten[Z];
  envt = Interpolation[Z, InterpolationOrder → 0];

  p = ninit;
  Do[nextp = nt[k, p];
   (*If[nextp<minPop〚i,l〛,minPop〚i,l〛=nextp];*)
   If[nextp < ((1 / α) * 10 ^ -3), extTime〚i, (*j,m,*) l〛 = k; Break[],
    p = nextp], {k, 2, tmax}],

  {l, 1, maxsims}, {i, 1, Length[γvals]}] // Timing

saveto = "/SAVETO/";
Export[saveto <> "model3.3.2_" <> species <> "_" <> when <> "_ext1.m", extTime, "MX"]
(*Export[
 saveto<>"model3.3.3_minPop_"<>species<>"_"<>when<>"_ext1.m",minPop,"MX"]*)
```